

University of the Armed Forces

DEPARTMENT OF COMPUTER SCIENCE

Object-Oriented Programming

HW04ProjectDefinition

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NRC: 36862

PROJECT DEFINITION

FlowDesk

Work Order and Cash Management System

Runtime Dynamics

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1. Project Information

Software name: FlowDesk — Order and Cash Management System for Small Businesses.

Development company: Runtime Dynamics.

Development team: Cristopher Lanchimba, Cristopher Guañuna, and Julián Jaramillo.

Initial context: a print shop that manages custom orders, deliveries, and payments.

Initial Context: A print shop that needs to organize customer information, products, services, work orders, payments, and cash movements.

2. Problem Statement

The print shop faces difficulties organizing and tracking work orders, deliveries, payments, income, and expenses. Customer information and order details can be difficult to find, making it harder to identify pending orders, check their status, and verify outstanding balances. Financial movements are also difficult to review without a centralized record.

FlowDesk aims to centralize customer information, product and service records, work orders, payments, and cash movements. The system will help track order progress, calculate totals and outstanding balances, manage financial transactions, and generate reports to improve the organization of daily operations.

3. Product Overview and Conditions

FlowDesk is designed to organize the print shop's daily operations. It will allow employees to register and search for customers using their personal and contact information, including identification numbers and addresses. Customer records will be associated with their work orders.

The system will manage products and services, including their names, descriptions, categories, prices, and quantities where applicable. Categories will help organize the products and services offered by the print shop.

Each work order will contain information about the requested work, its associated customer, the products or services involved, and the corresponding details, including descriptions, quantities, unit prices, and totals. Orders will be searchable by identification code and will have one of the following statuses: Waiting, Approved, In Progress, Completed, or Delivered.

Payments will be registered with their amounts, dates, payment methods, and associated work orders. The system will support cash payments and bank transfers, allowing employees to track payments received and outstanding balances.

Cash management will include recording income and expenses, identifying the type and reason for each movement, and calculating total income, expenses, and the resulting balance. Payments may be associated with the corresponding cash movements when applicable.

A dashboard will display the number of work orders according to their status and allow employees to filter orders. The system will also generate daily, weekly, and monthly reports for reviewing work orders and financial activity.

Users will be able to log in and log out, with credential validation to control access to the system. Dates will be recorded using day, month, and year information. The system will distinguish between order statuses, payment methods, movement types, and report types to organize information consistently.

The project will follow an object-oriented programming approach and prioritize organized information, clear relationships between records, and straightforward management of the print shop's operations.

4. Background

Developing FlowDesk requires basic programming knowledge, object-oriented programming, and software design. The development team must understand how to organize information, define operations, and establish relationships between different types of records.

Knowledge of information management is necessary to register, search, update, and maintain consistent customer, product, service, and order information. The team must also understand how to connect work orders with their details, payments, and related financial movements.

Basic user interface design is needed to present order information, dashboards, and management functions clearly. Authentication concepts are also necessary to manage access through login, logout, and credential validation.

Finally, the team must understand the print shop's workflow, including order creation, progress tracking, delivery, payment registration, and the management of income and expenses. Basic financial concepts are required to calculate totals and outstanding balances and to prepare daily, weekly, and monthly reports.